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Studierichting: Informatica
Oefeningenreeks 7B nr. 7.2

$f : \Omega \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}^2 : (x,y) \longmapsto (e^{xy}, \ln \frac{x}{y})$

$$Df(x,y) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} ye^{xy} & xe^{xy} \\ \frac{1}{x} & -\frac{1}{y} \end{pmatrix}$$

$\det Df^{-1}(x,y) = (ye^{xy})(-\frac{1}{y}) - (xe^{xy})(\frac{1}{x}) = -e^{xy} - e^{xy} = -2e^{xy}$

$e^{xy} > 0$ voor alle $(x,y) \in \Omega \Rightarrow$ inverteerbaar.

afgeleide van de inverse:

$$Df^{-1}(a,b) = \frac{1}{-2e^{xy}} = \begin{pmatrix} -\frac{1}{y} & -xe^{xy} \\ -\frac{1}{x} & ye^{xy} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2ye^{xy}} & \frac{x}{2} \\ \frac{1}{2xe^{xy}} & -\frac{y}{2} \end{pmatrix}$$

References